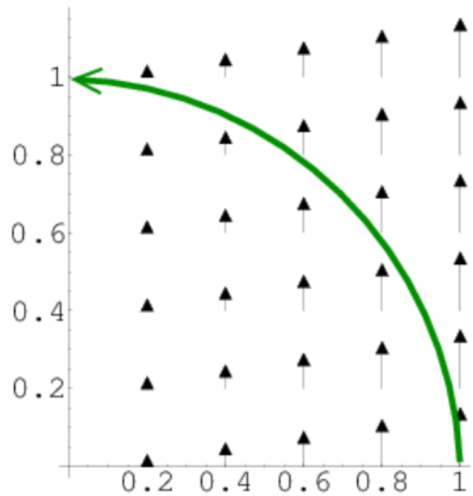
Week 1 Discussion Solutions

Math Review

1. Given the force $\vec{F}(x,y) = (0,x)$, calculate the work done by the force field on a particle that moves counterclockwise in a quarter unit circle in the first quadrant.



Solution:

$$W = \int \vec{F} \cdot d\vec{s} \tag{2}$$

The easiest way of evaluating this integral is to switch use polar coordinates: $x = r\cos(\theta)$, $y = r\sin(\theta)$. In this case, r is always equal to 1, so we have $\vec{F}(\theta) = (0, \cos(\theta))$. The path $\vec{s} = (x, y)$ can be written as $\vec{s} = (r\cos(\theta), (r\sin(\theta))$, or simply:

$$\vec{s} = (\cos(\theta), \sin(\theta)) \tag{3}$$

Taking the derivative:

$$\frac{d\vec{s}}{d\theta} = (-\sin(\theta), \cos(\theta)) \tag{4}$$

So $d\vec{s} = d\theta(-\sin(\theta), \cos(\theta))$ Now we can evaluate the integral:

$$W = \int \vec{F} \cdot d\vec{s} = \int_0^{\pi/2} (0, \cos(\theta)) \cdot (-\sin(\theta), \cos(\theta)) d\theta \tag{5}$$

$$= \int_0^{\pi/2} \cos^2(\theta) d\theta = 0.5 \int_0^{\pi/2} (1 + \cos(2\theta)) \tag{6}$$

$$= 1/2[\theta + \sin(2\theta)/2]_0^{\pi/2} = \pi/4 \tag{7}$$

Work is path depended (for example it's 0 along the axis) which makes sense as the field isn't conservative (its curl is nonzero).

2. Calculate the surface integral of $\vec{v} = 2xz\hat{x} + (x+2)\hat{y} + y(z^2-3)\hat{z}$ over five sides (excluding the bottom) a cubical box of size 2. Let upward and outward be the positive direction.

   Solution: Taking the sides one at a time:

   (i) $x = 2$, $d\vec{a} = dydz\hat{x}$, and $\vec{v} \cdot d\vec{a} = 2xzdydz = 4zdydz$, so

$$\int \vec{v} \cdot d\vec{a} = 4 \int_0^2 dy \int_0^2 zdz = 16. \tag{8}$$

(ii) $x = 0$, $d\vec{a} = -dydz\hat{x}$, and $v \cdot d\vec{a} = -2xzdydz = 0$, so

$$\int \vec{v} \cdot d\vec{a} = 0. \tag{9}$$

(iii) $y = 2$, $d\vec{a} = dxdz\hat{y}$, and $\vec{v} \cdot d\vec{a} = (x+2)dxdz$, so

$$\int \vec{v} \cdot d\vec{a} = \int_0^2 (x+2)dx \int_0^2 dz = 12. \tag{10}$$

(iv) $y = 0$, $d\vec{a} = -dxdz\hat{y}$, and $\vec{v} \cdot d\vec{a} = -(x+2)dxdz$, so

$$\int \vec{v} \cdot d\vec{a} = -\int_0^2 (x+2)dx \int_0^2 dz = -12. \tag{11}$$

(v) $z = 2$, $d\vec{a} = dxdy\hat{z}$, and $\vec{v} \cdot d\vec{a} = y(z^2-3)dxdy = ydxdy$, so

$$\int \vec{v} \cdot d\vec{a} = \int_0^2 dx \int_0^2 ydy = 4. \tag{12}$$

Total flux = 16 + 0 + 12 - 12 + 4 = 20

3. (a) Choose parameterization $\gamma(t) = (t, y_0 t)$ for $t \in [0,1]$. $\gamma'(t) = (1, y_0)$, so the integral is

$$W = \int_0^1 (-k, \frac{-ky_0 t}{y_0}) \cdot (1, y_0)dt = -k\left(1 + \frac{1}{2}y_0\right) \tag{13}$$

(b) $\gamma(t) = (\sin t, y_0 \sin t)$ for $t \in [0, \pi/2] \to \gamma'(t) = (\cos t, y_0 \cos t)$, so the integral is

$$W = \int_0^{\pi/2} (-k, \frac{-ky_0 \sin t}{y_0}) \cdot (\cos t, y_0 \cos t)dt = -k\left(1 + \frac{1}{2}y_0\right) \tag{14}$$

[Note: This vector field is conservative, i.e., $\vec{F} = -\vec{\nabla}U$, where $U = kx + \frac{ky^2}{2y_0}$. Therefore any path between these two points should give the same answer.]

Thermodynamics

4. Hot water. To remove the lid, we want the metal to expand more than the glass does, which can be achieved by increasing the temperature.

5. The volume expansion formula is

$$V(T) = V_0 + \Delta V = V_0(1 + \beta(\Delta T)), \tag{15}$$

where β is the volume expansion coefficient. Using the linear expansion formula, show $\beta = 3\alpha$, where α is the linear expansion coefficient in each linear direction (assume the material expands homogeneously).

Solution: The equation for linear expansion is given by

$$L(T) = L_0(1 + \alpha\Delta T), \tag{16}$$

and so the volume expansion formula may be rewritten

$$\begin{aligned} V(T) = L(T)^3 &= L_0(1+\alpha\Delta T) \times L_0(1+\alpha\Delta T) \times L_0(1+\alpha\Delta T) \\ &= V_0(1 + 3\alpha\Delta T + 3\alpha^2\Delta T^2 + \alpha^3\Delta T^3). \end{aligned} \tag{17}$$

Matching terms, and ignoring terms order ΔT^2 and higher, we obtain the result.

6. We start with the linear expansion of the length $L(T) = L_0(1 + \alpha_L \Delta T)$ and the radius $R(T) = R_0(1 + \alpha_R \Delta T)$. Then we use these in the formula for volume and only keep terms which are linear in ΔT, because $\frac{\Delta L}{L_0} = \alpha_L \Delta T \ll 1$ (and similarly $\frac{\Delta R}{R_0} = \alpha_R \Delta T \ll 1$).

$$\begin{aligned} V(T) &= \pi [R(T)]^2 [L(T)] \\ &= \pi R_0^2 L_0 (1 + \alpha_R \Delta T)^2 (1 + \alpha_L \Delta T) \\ &= V_0 (1 + 2\alpha_R \Delta T)(1 + \alpha_L \Delta T) \\ &= V_0 [1 + (2\alpha_R + \alpha_L) \Delta T] \\ &= V_0 (1 + \beta \Delta T) \end{aligned}$$

So, our result is

$$\beta = 2\alpha_R + \alpha_L. \tag{18}$$

7. There is a lot of extra information in this problem, designed to trick you into thinking it's a volume problem. However, the simplest way to do this is just using one dimension. Ultimately, the solution is very simple:

$$L_0 \alpha \Delta T = L_0 \frac{\beta}{3} \Delta T = 50\,\mathrm{m} \times \frac{1}{3} \times 2.10 \times 10^{-4}\,\mathrm{K}^{-1} \times 0.5\,\mathrm{K} = 0.175\,\mathrm{cm}. \tag{19}$$

8. (a) In order for $f(v)$ to be a valid probability density, we must have

$$\int_0^\infty f(v)dv = 1 \implies \int_0^{v_0} Av^2 dv = 1 \implies A = \frac{3}{v_0^3}$$

(b)

$$\bar{v} = \int_0^{v_0} v f(v) dv = \int_0^{v_0} v \left(\frac{3}{v_0^3}\right) v^2 dv = \frac{3}{4} v_0$$

(c) $v_\mathrm{p} = v_0$. Taking a derivative will not help you find this because $f(v)$ is discontinuous at v_0. The probability of measuring the speed of an individual gas molecule and obtaining any particular speed v' is 0; it is only meaningful to talk about the probability of measuring a speed v' in $(v', v' + dv) = f(v)dv$.

(d) $v_\mathrm{RMS} = \sqrt{\langle v^2 \rangle}$. So

$$\langle v^2 \rangle = \int_0^{v_0} v^2 f(v) dv = \int_0^{v_0} v^2 \left(\frac{3}{v_0^3}\right) v^2 dv = \frac{3}{5} v_0^2 \implies v_{rms} = v_0 \sqrt{\frac{3}{5}}$$

(e) It is easier to calculate the variance and then the standard deviation. To calculate the variance, we need to know the average squared distance from the mean, so we have

$$\sigma^2 = \int_0^{v_0} \left(v - \frac{3}{4} v_0\right)^2 f(v) dv = \frac{1}{v_0^3} \int_0^{v_0} \left(3v^4 - \frac{9}{2} v_0 v^3 + \frac{27}{16} v_0^2 v^2\right)$$

$$\sigma^2 = v_0^2 \left(\frac{3}{5} - \frac{9}{8} + \frac{9}{16}\right) = \frac{3}{80} v_0^2 \rightarrow \sigma = \frac{3}{4\sqrt{5}} v_0$$

[You can also use $\sigma^2 = \langle v^2 \rangle - \langle v \rangle^2$. This could be a good way to introduce the formula, as it saves a rather tedious calculation.]